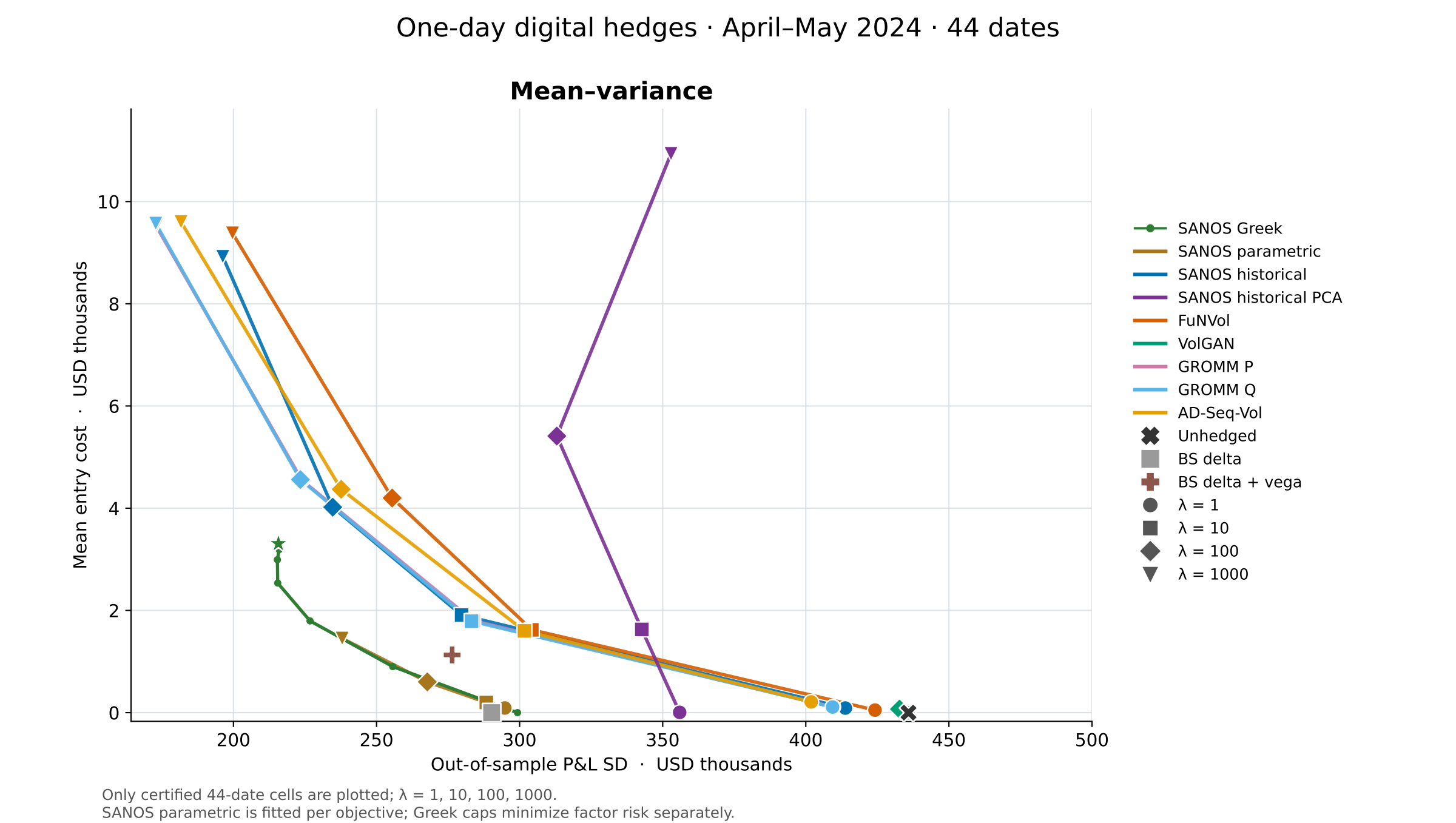
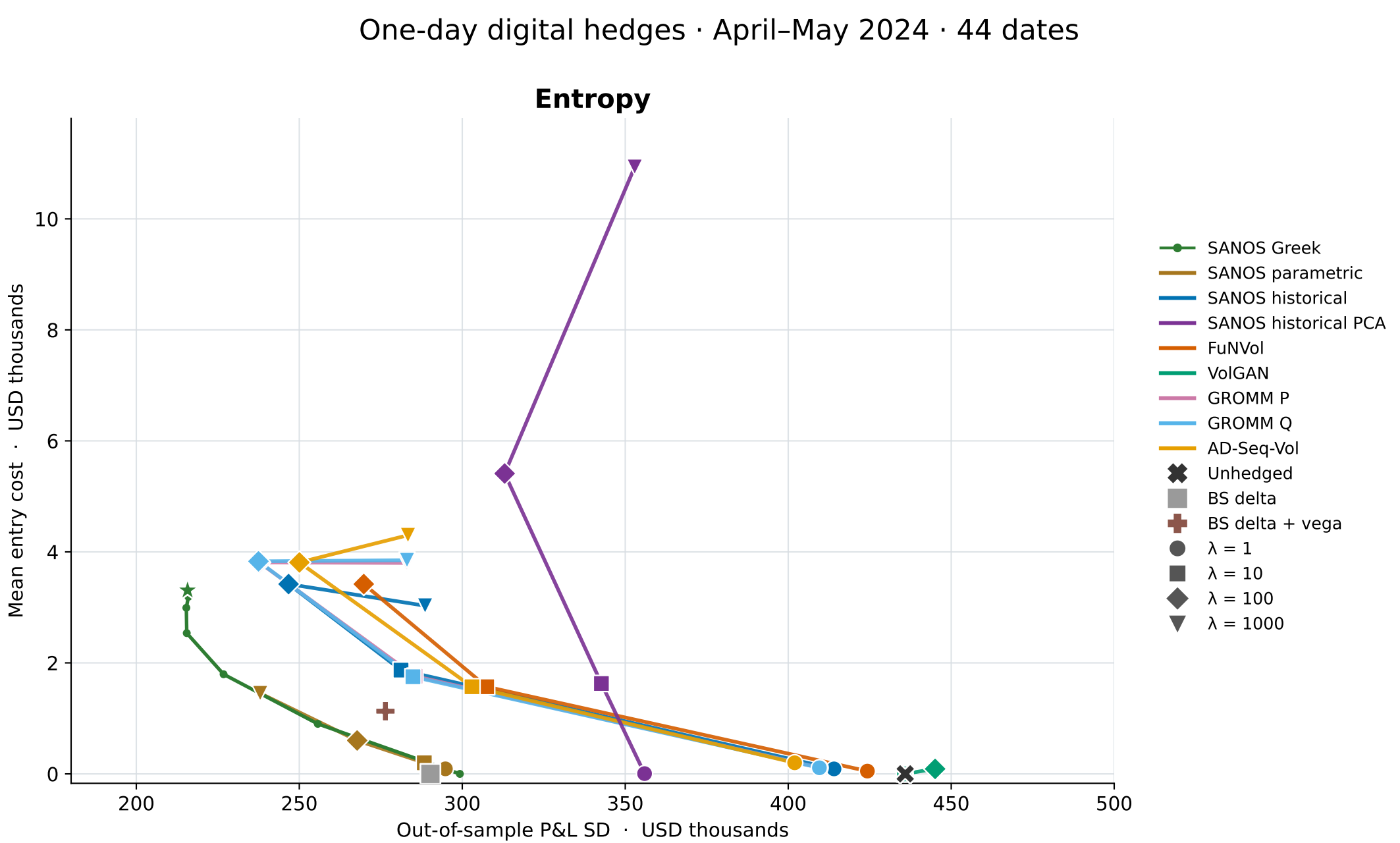
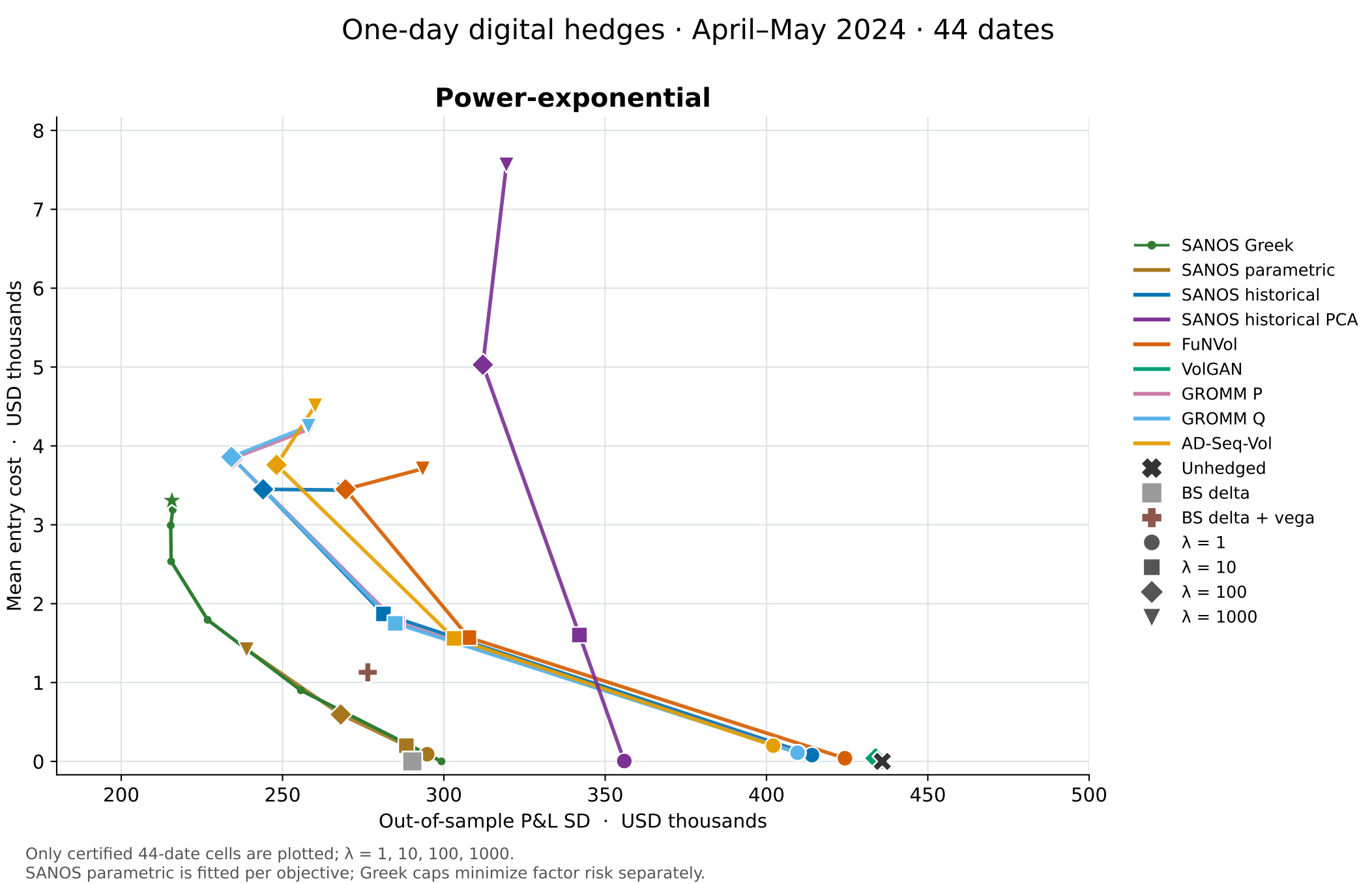
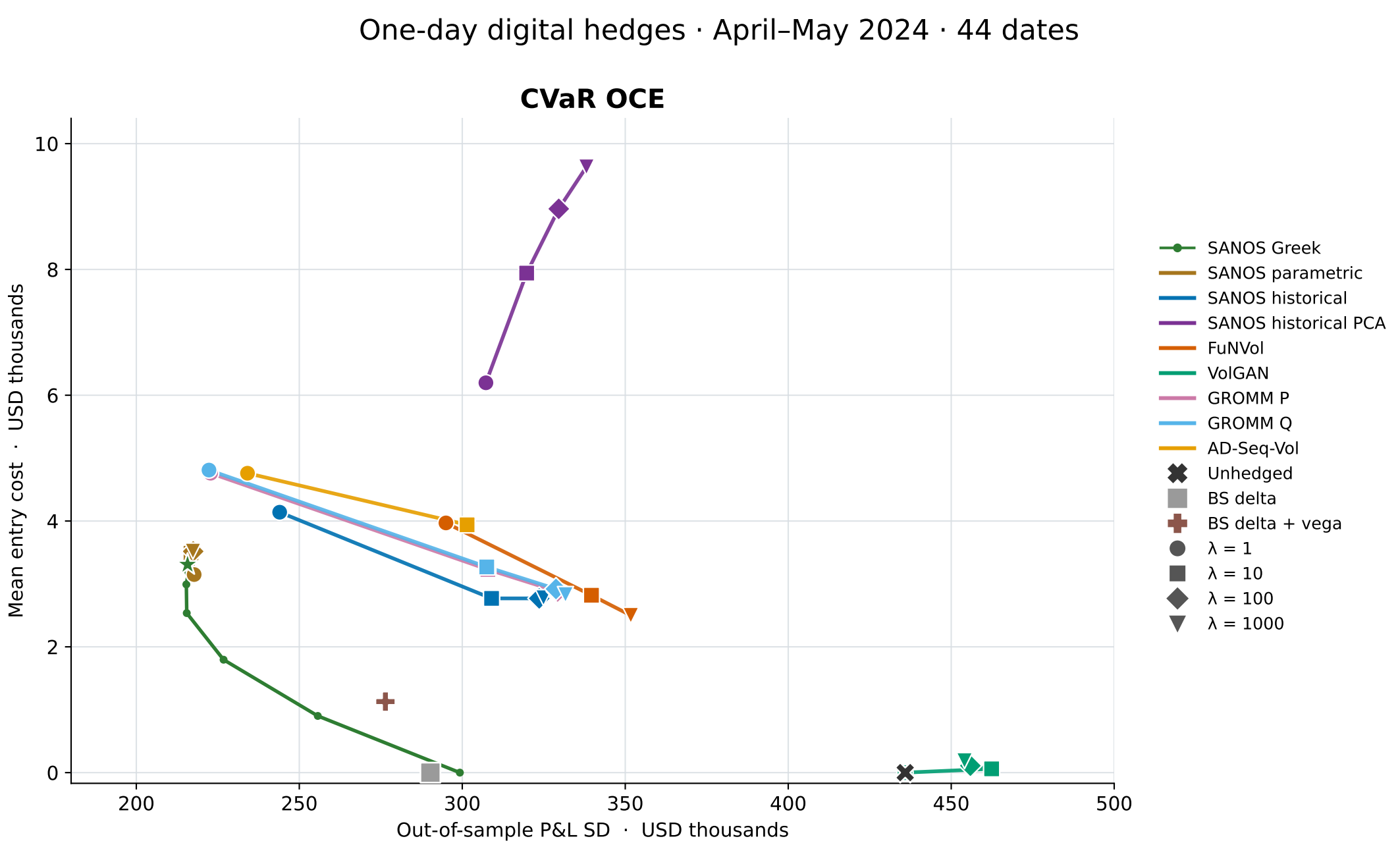

One-day digital hedges · April–May 2024 · 44 dates
Mean–variance
Mean entry cost · USD thousands
Out-of-sample P&L SD · USD thousands
0
2
4
6
8
10
200
250
300
350
400
450
500
SANOS Greek
SANOS parametric
SANOS historical
SANOS historical PCA
FuNVol
VolGAN
GROMM P
GROMM Q
AD-Seq-Vol
Unhedged
BS delta
BS delta + vega
λ = 1
λ = 10
λ = 100
λ = 1000
Only certified 44-date cells are plotted; λ = 1, 10, 100, 1000.
SANOS parametric is fitted per objective; Greek caps minimize factor risk separately.



Only certified 44-date cells are plotted; λ = 1, 10, 100, 1000.
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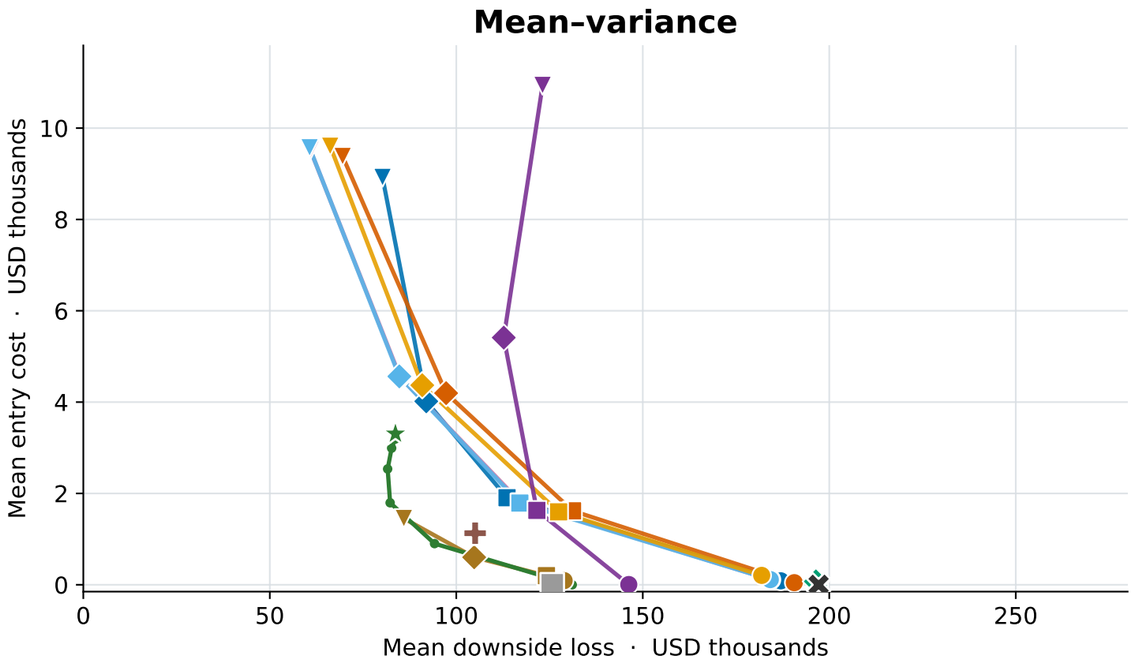
Only certified 44-date cells are plotted; λ = 1, 10, 100, 1000.
SANOS parametric is fitted per objective; Greek caps minimize factor risk separately.

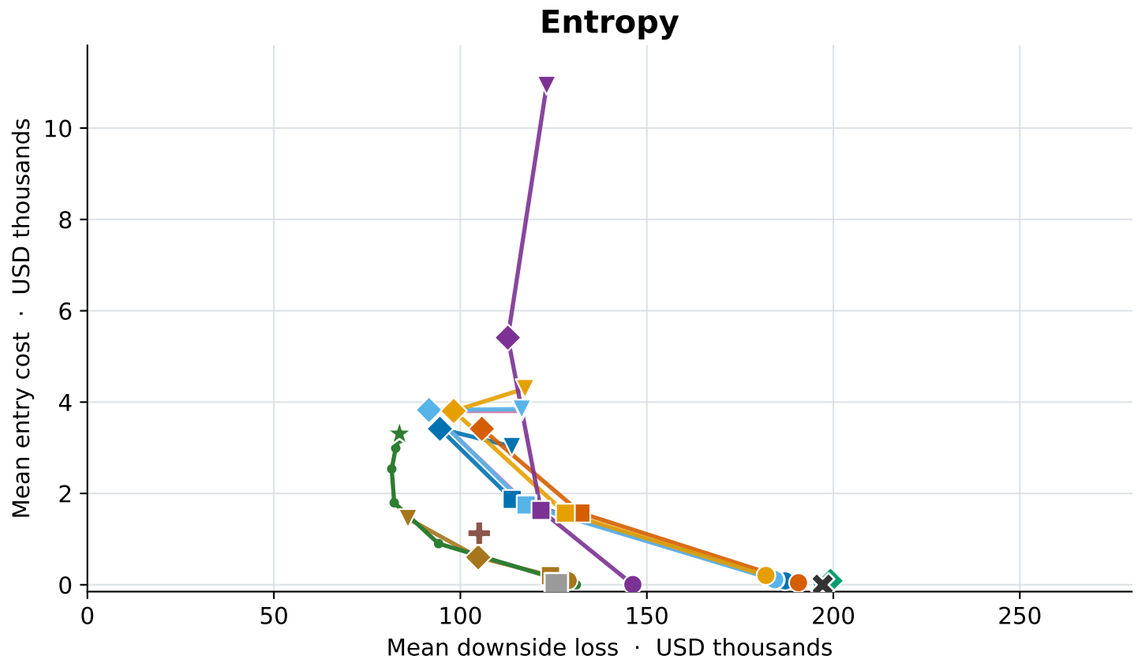
Coverage of the 44-date objective grid · snapshot 29 Oct 2026, 07:47 UK

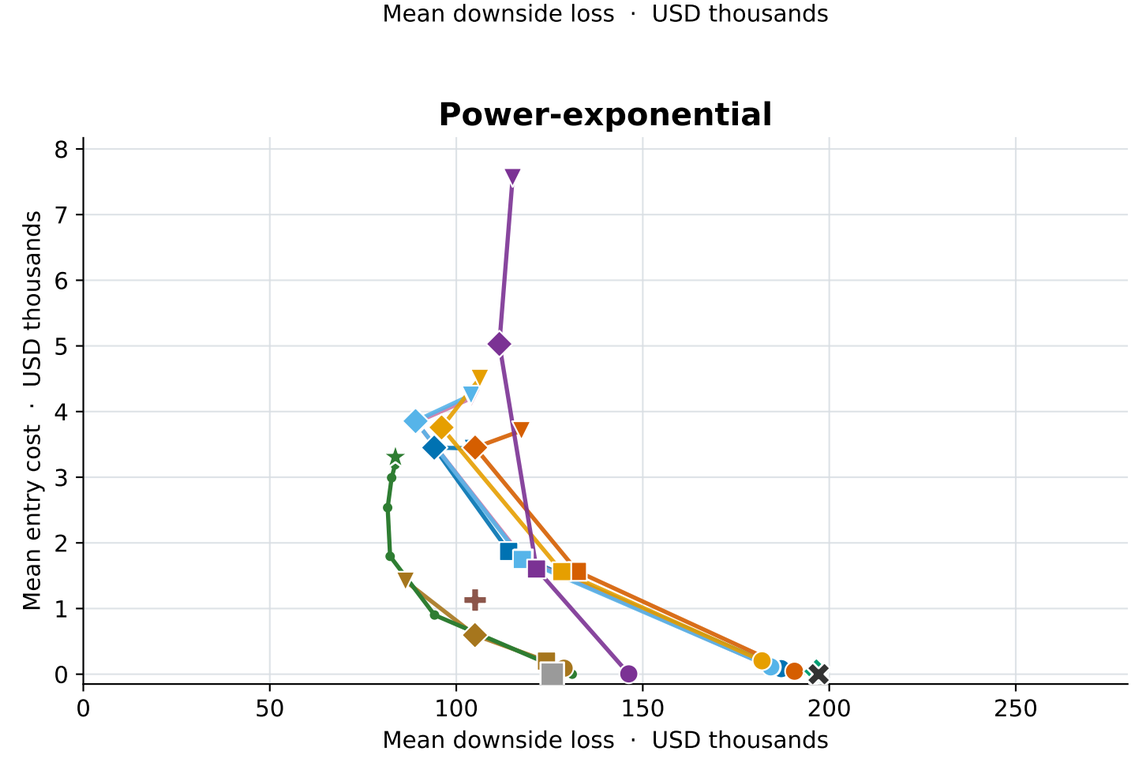
Objective	λ	SANOS param.	SANOS hist.	SANOS hist. PCA	FuNVol	VolGAN	GROMM P	GROMM Q	AD-Seq-Vol	AD-Seq-Vol-FT
mean-variance	1	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Not run
mean-variance	10	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Complete	1/44
mean-variance	100	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Not run
mean-variance	1000	Complete	Complete	Complete	Complete	Failed	Complete	Complete	Complete	Not run
entropy	1	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Not run
entropy	10	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Not run
entropy	100	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Not run
entropy	1000	Complete	Complete	Complete	Failed	Failed	Complete	Complete	Complete	Not run
power-exponential	1	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Not run
power-exponential	10	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Not run
power-exponential	100	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Not run
power-exponential	1000	Complete	Complete	Complete	Complete	Failed	Complete	Complete	Complete	Not run
cvar	1	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Not run
cvar	10	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Not run
cvar	100	Complete	Complete	Complete	Failed	Complete	Complete	Complete	23/44	Not run
cvar	1000	Complete	Complete	Complete	Complete	Complete	Not run	Complete	Not run	Not run

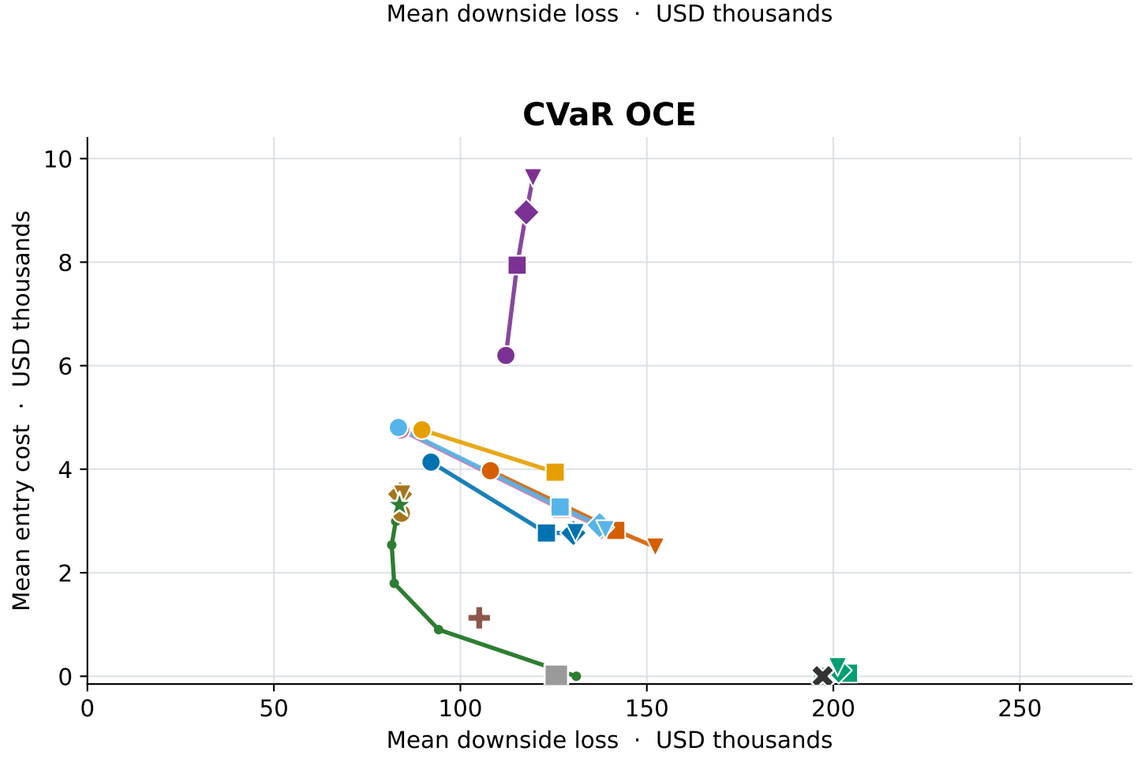
Complete = certified and plotted; n/44 = partial, not plotted; Failed = certification failed; Cap-limited = position limit reached; Not run = no saved dates.

Entry cost versus mean downside loss E[max(−PnL, 0)] · one-day digital hedges









SANOS Greek
SANOS parametric
SANOS historical
SANOS historical PCA
FuNVol
VolGAN
GROMM P
GROMM Q
AD-Seq-Vol
Unhedged
BS delta
BS delta + vega
λ = 1
λ = 10
λ = 100
λ = 1000

y 2024 dates; ten targets. Loss uses costed out-of-sample long-claim P&L. Risk aversions λ = 1, 10, 100, 1000 where complete. Greek caps optimize factor risk separately; SANOS parametric appears only where independently fitted for the plotte